Charging System Update

- Initially:
  a. Solar Panel
  b. MPPT charge controller
  c. Rover battery
- Added:
  a. Base station battery 12V 200Ah
  b. DC-DC charge controller
  c. Multi tap connector
- Tested modular systems & connections

Next Steps: test full system with contacts and solar panel







POC: Lizeth Montemayor (lm945), Antranig Baghdassarian (ab2864), Casey Kashishian (ck677)

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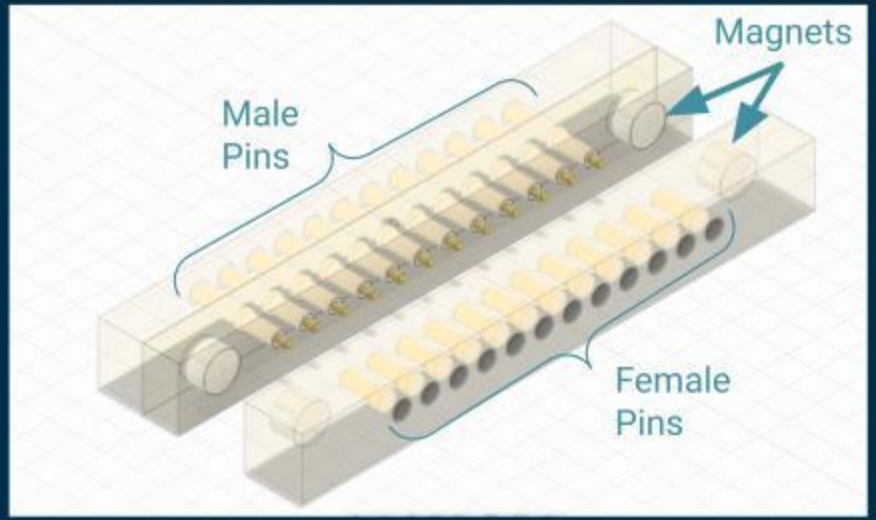
Charging Contacts

- Goal: Connect Rover battery to charger (inside the base station) autonomously, **safely**!

Previous Prototype:



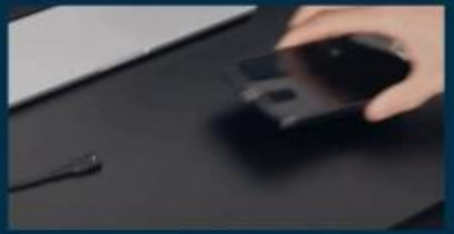
Current Design:



How we achieve autonomy:



End Result:



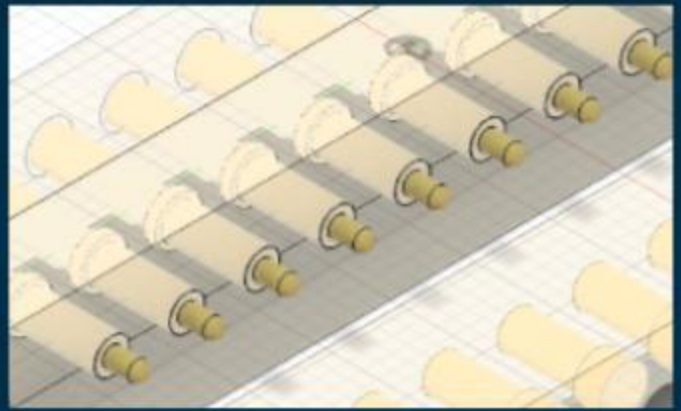
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Charging Contacts: How it connects

- Goal: Connect Rover battery to charger (inside the base station) autonomously, **safely**!

Recall:



Called Pogo pins! (like the pogo stick!) →

Anatomy:



Can compress to always allow for solid connection!



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Charging Contacts: Next Steps

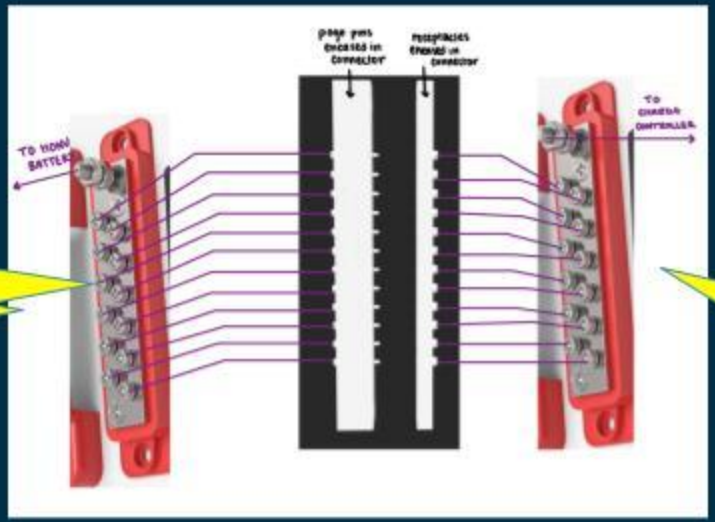
1. Finish wiring connectors to bus bar!
2. Stress test connection (How good is the 3D printed connector?)
3. Figure out enclosure (to prevent water getting in)
4. Implement onto base station and rover!
5. Celebrate



Honu (our rover):



Couldn't find a good pic



Base Station (w/o Liz & Trin):



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